EX NO-10
DATE-

INTERNETWORKING WITH ROUTERS IN CISCO PACKET TRACER SIMULATOR

AIM: To design and configure an internetwork using routers, wireless router, DHCP server and Internet Cloud in Cisco Packet tracer, and to verify connectivity between the network devices.

REQUIREMENTS:

* Cisco packet Tracer
* Router
* Wireless Router
* PCs
* Laptop
* Cable Modem
* Internet Cloud
* Cisco.com Server
* Copper straight-Through cable
* Coaxial cable

THEORY: Internetworking is the process of connecting different networks using networking devices such as routers and wireless routers. A router forwards data packets between different IP networks, while a wireless router provides wired and wireless connectivity to end devices.

DHCP (Dynamic host Configuration Protocol) automatically assigns IP addresses, subnet masks, default gateways and DNS server information to network devices. DNS (Domain Name System) is used to translate domain names into IP addresses.

In this experiment, two different networks are connected using a router. A wireless network is also created using a wireless router. The PC and laptop obtain their IP configuration using DHCP and an Inter cloud and Cisco.com server are used to demonstrate Internet connectivity.

NETWORK CONFIGURATION:

Router Configuration:

Device	Interface	IP address	Subnet Mask
Router 1	FastEthernet 0/0	192.168.10.1	255.255.255.0
Router 2	FastEthernet 0/1	192.168.20.1	255.255.255.0

PC configuration:

Device	Interface	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Wireless Network Configuration:

Device	Interface	IP address	Subnet Mask	Gateway
Wireless Router	LAN	192.168.0.1	255.255.255.0	-
PC	Ethernet 0	DHCP	255.255.255.0	192.168.0.
Laptop	Wireless 0	DHCP	255.255.255.0	192.168.0.
Cisco.com Server	Ethernet 0	208.67.220.220	255.255.255.0	208.67.22

PROCEDURE:

Part 1: Creating the Network Topology:

1) Open Cisco Packet Tracer and create a new workspace
2) Add the required routers, wireless router, PCs, laptop, cable modem, Internet cloud and Cisco.com server.
3) Connect PC0 and PC1 to the ~~server~~ router using CV straight through cables.
4) Connect the wireless router to the cable modem
5) Connect the cable modem to the Internet cloud using a coaxial cable.
6) Connect the Internet cloud to the Cisco.com

Part-2: Configuring the Routers:

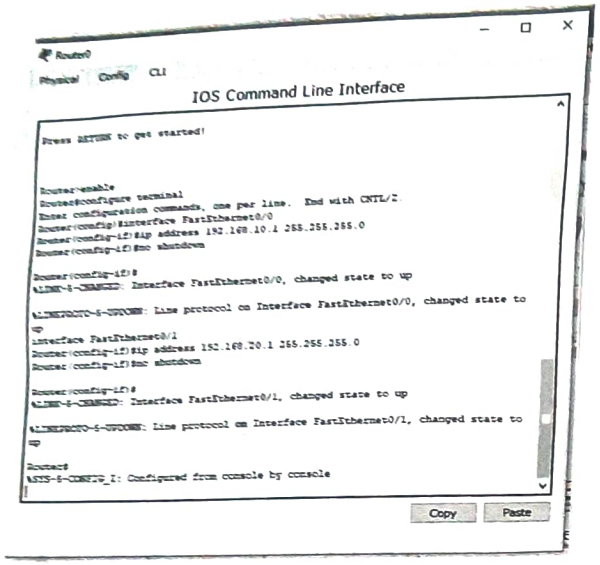
1) Open the CLI of Router1
2) Enter the privileged mode using enable.
3) Enter the configuration mode using configure terminal
4) Configure Fast Ethernet 0/0 with IP address 192.168.10.1 and subnet mask 255.255.255.0
5) Configure Fast Ethernet 0/1 with IP address 192.168.20.1 and subnet mask 255.255.255.0
6) Use the no shutdown command to activate both interface.

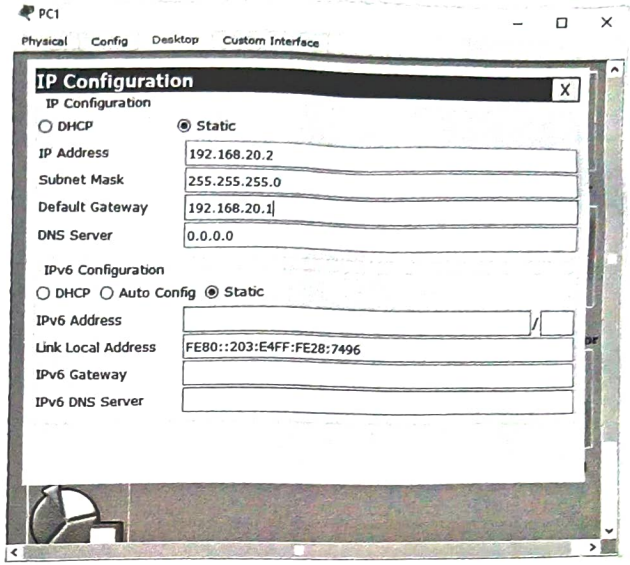
Part-3: Configuring the PCs.

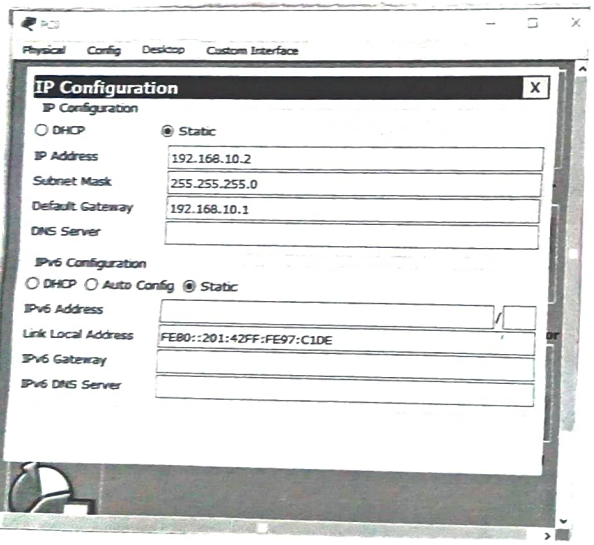
1) Open the IP configuration window of PC0.
2) Assign the IP address 192.168.10.2
3) Set the default gateway as 192.168.10.1
4) Configure PC1 with IP address 192.168.20.2
5) Set the default gateway as 192.168.20.1
6) Test communication between PC0 and PC1 using the ping command (or Simple PDU).

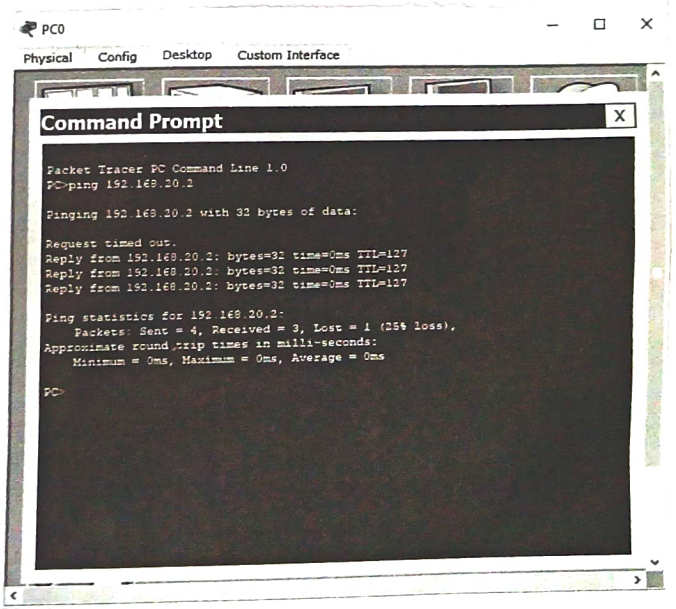
Part 4 - Configure the wireless Routing:

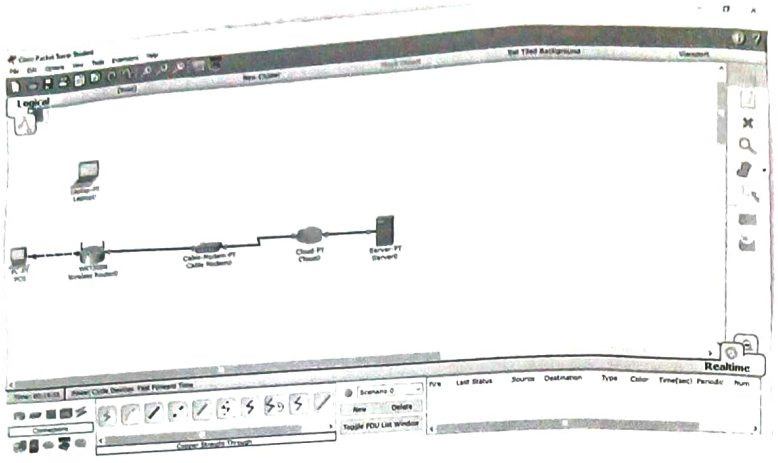
1) Open the wireless router configuration window
2) Go to GUI → Wireless.
3) Set the SSID as Home Network.
4) Go to the setup tab.
5) Enable the DHCP Server.
6) Configure the DNS Server as 208.67.220.220
7) Save the configuration.

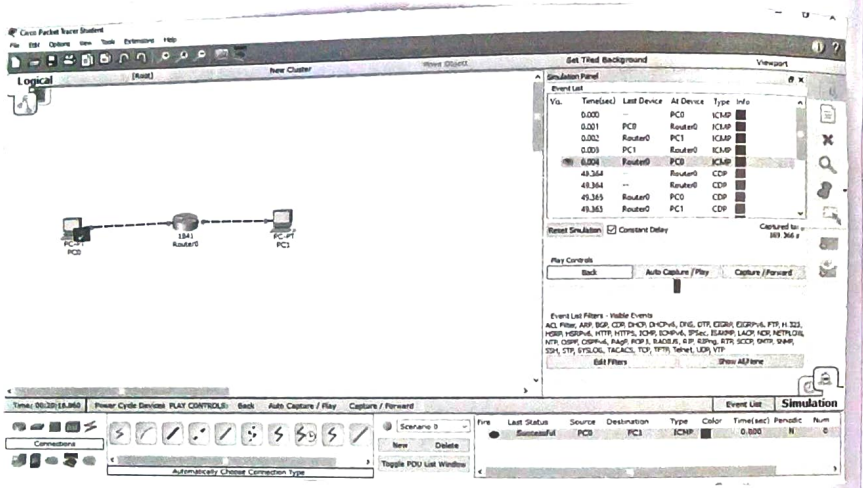

Router0
Physical
Config
CLI
IOS Command Line Interface
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste


PC1
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
IP Address
192.168.20.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::203:E4FF:FE28:7496
IPv6 Gateway
IPv6 DNS Server


PC2
Physical
Config
Desktop
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
IP Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::201:42FF:FE97:C1DE
IPv6 Gateway
IPv6 DNS Server


PC0
Physical
Config
Desktop
Custom Interface
Command Prompt
Packet Tracer PC Command Line 1.0
PC>ping 192.168.20.2
Pinging 192.168.20.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Reply from 192.168.20.2: bytes=32 time=0ms TTL=127
Ping statistics for 192.168.20.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
PC>





Part 5 - Configuring the Laptop:

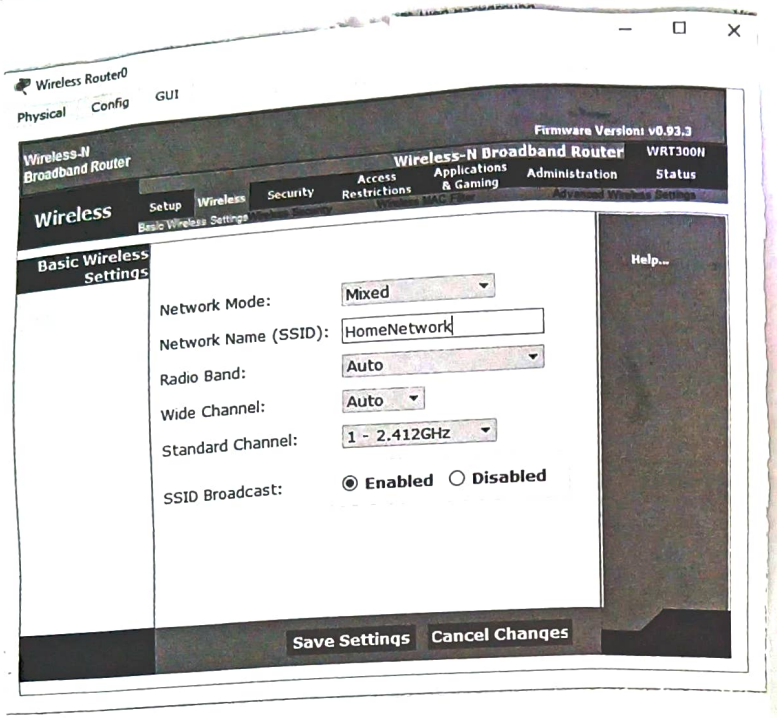
1) Open the laptop configuration window.
2) Install the wireless WPC 300N module if required
3) Open Desktop → PC Wireless.
4) Search for the Wireless network Home Network.
5) Select the network and connect the laptop.
6) Configure the laptop to obtain its IP address automatically using DHCP.

Part 6 - Configuring the PC with DHCP:

1) Open the PC and Select Desktop → IP Configuration
2) Select DHCP
3) The PC receives an IP address automatically from the wireless router.
4) Use ipconfig /all to verify the IP address, subnet mask, gateway and DNS Server.

Part 7 - Configuring the cisco.com Server:

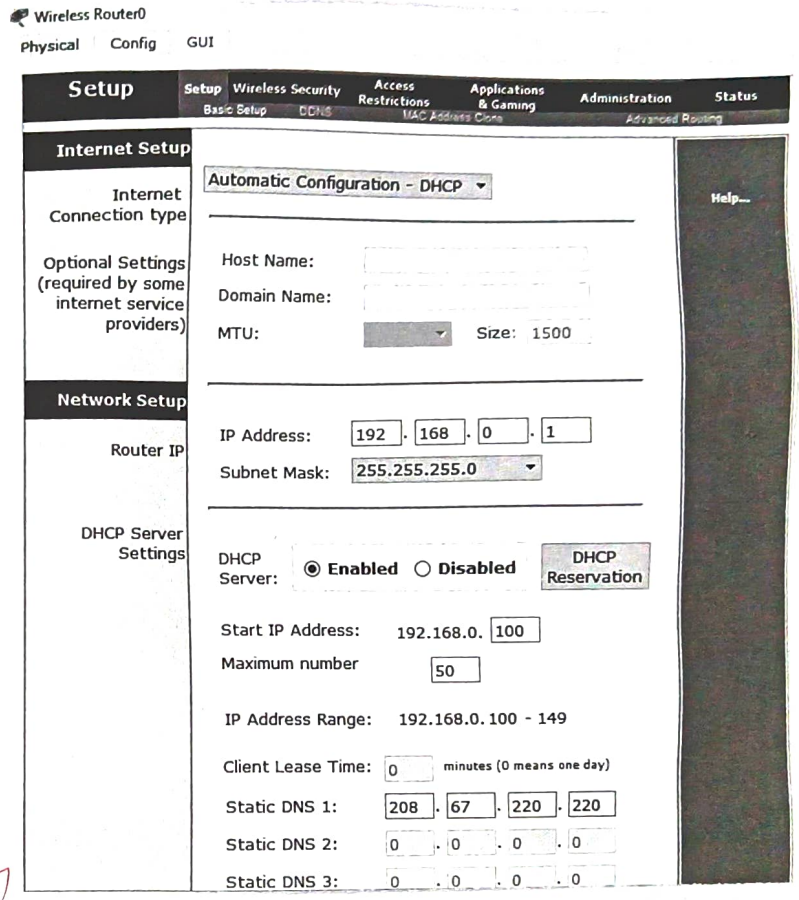
1) Open the cisco.com Server and select the services tab.
2) Enable the DHCP Service
3) Create a DHCP pool named DHCP pool.
4) Configure the required gateway, DNS server, starting IP address and subnet mask.
5) Enable the DNS service.
6) Add the DNS record:
   - Name: Cisco.com
   - Type: A Record
   - Address: 208.67.220.220.
7) Configure the server with the static IP address 208.67.220.220.



Part 8: Testing the Network:

1) Open the PC command prompt.
2) Use ipconfig/release and ipconfig/renew to refresh the DHCP configuration.
3) Verify that the PC receives an IP address in the 192.168.0.x range.
4) Verify that successful replies are receiv
5) Send a simple PDU between the required devices and observe the successful transmiss

OBSERVATION: The internetwork was successfully created by connecting different network using a router and wireless router. The router was configure with two different networks, 192.168.10.0/24 and 192.168.20.0/24, allowing communicatio between PC0 and PC1. The wireless router w configured with the SSID. The wireless router Home Network, and DHCP was used to automatically assign IP configuration to the connected devices. The Cisco.com server was configured with DHCP and DNS services. Connecti was verified using ping and simple PDU and successful communication was observed.



RESULT: Thus, an internetwork using a router, wireless router, DHCP server and Internet Cloud was successfully designed and configured in Cisco packet Tracer, and Connectivity between the network devices was successfully verified.